

MARATHON®



SPECIFICATIONS

Top Terminal Batteries

From the World Leader in VRLA Battery Technology

Designed for durability in telecommunications and electric utility applications, the top terminal **MARATHON®** series provides high performance and reliability in long duration discharge applications. The **MARATHON®** family of batteries highlights another example of GNB Industrial Power's extensive experience and world wide leadership in VRLA technology.

"Designed-in" Quality Manufacturing

Quality manufacturing processes for the **MARATHON®** series batteries incorporate the industry's most advanced technologies including: an automated helium leak detection system, a computer controlled "fill by weight" acid filler, and a temperature controlled water bath formation process. Each and every unit is capacity tested.

High Performance MARATHON® Series Features

- Standard: Reinforced polypropylene container and cover.
- Optional: Flame-retardant reinforced container and cover compliant with UL94 V-0.
- Integrated Flash-arrester ultrasonically welded into cover.
- Patented "Diamond Side-Wall" design to maintain structural integrity in higher operating temperatures.
- Heat sealed case-to-cover bond to ensure a leak proof seal.
- High-Compression Absorbent Glass Mat (AGM) technology for greater than 99% recombination efficiency.
- Tin, calcium, silver, lead positive plate design for maximum service float life; 10 year design life @ 25°C (77°F).
- Heavy duty copper alloy terminals for ease of assembly and reduced maintenance.
- Reliable one-way, self-resealing safety vents.
- Multicell design for faster installation and reduced maintenance.
- Horizontal or vertical operation.
- Removable carry handles for ease of installation.

Applications

MARATHON® series batteries incorporate GNB's advanced VRLA technology designed for long life and high performance in:

Telecommunications

- Wireless
- Broadband
- Signaling
- Distributed Power
- Customer Premise

Electric Utility

- Switchgear Control Power
- Communications
- SCADA

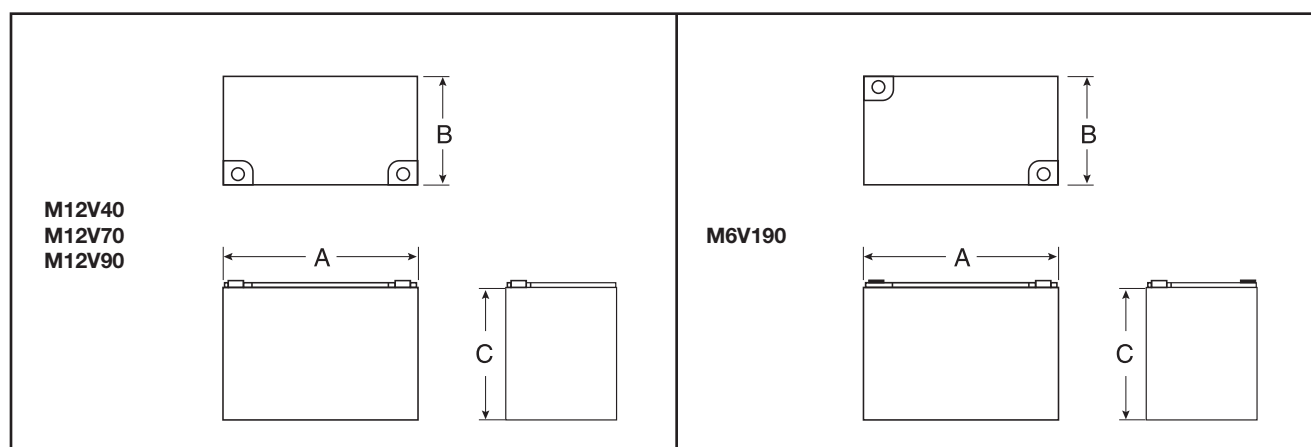


MARATHON® Specifications

Model Number [#]	Voltage	Capacity (AH)		Nominal Dimensions						Nominal Weight	
		8 hr To 1.75 VPC @ 25°C	10 hr To 1.80 VPC @ 20°C	Inches			Millimeters				
				A	B	*C	A	B	*C	lbs.	Kg
M12V40 [#]	12	40	37	7.81	6.58	7.01	198	167	178	39	17.8
M12V70 [#]	12	72	71	10.25	6.85	8.80	260	174	224	61	27.8
M12V90 [#]	12	90	88	12.05	6.85	8.80	306	174	224	72	32.8
M6V190 [#]	6	190	186	12.05	6.85	8.80	306	174	224	74	33.5

* Bolt, washer, and connector typically increase height by 0.45 in. (11 mm)

Add suffix "F" to model number for flame retardant OPTION



MARATHON® Electrical Data

Model Number	Short Circuit Current (Amps)	Internal Resistance (mOhms)
M12V40	2341	5.3
M12V70	3271	3.7
M12V90	3365	3.7
M6V190	6343	1.0

Float Voltage & Charging

Constant Voltage charging is recommended

Recommended float voltage: 2.27 VPC @ 25°C (77°F)

Float Voltage Range: 2.25 to 2.30 VPC @ 25°C (77°F)

Equalize voltage: 2.35 VPC for 24 Hours

NOTE: Design and/or specifications subject to change without notice. If questions arise, contact your local GNB sales representative for clarification

MARATHON® Performance Specifications Amperes @ 25°C (77°F)

	Model Number	Time														
		24 hr	12 hr	10 hr	9 hr	8 hr	7 hr	6 hr	5 hr	4 hr	3 hr	2.5 hr	2 hr	1.5 hr	1 hr	0.5 hr
1.75 Final VPC	M12V40(F)	1.7	3.4	4.1	4.4	5.0	5.6	6.4	7.6	9.3	11.9	14.0	16.9	21.5	30.5	51.3
	M12V70(F)	3.2	6.2	7.4	8.2	9.1	10.1	11.6	13.4	16.2	20.6	24.0	28.9	36.8	51.6	90.8
	M12V90(F)	4.2	7.8	9.2	10.1	11.2	12.7	14.3	16.7	20.2	25.9	30.2	36.5	46.6	65.7	107.0
	M6V190(F)	9.1	16.6	19.5	21.4	23.7	26.7	30.6	35.9	43.7	56.0	65.6	79.5	102.0	144.9	246.0

	Model Number	Time														
		24 hr	12 hr	10 hr	9 hr	8 hr	7 hr	6 hr	5 hr	4 hr	3 hr	2.5 hr	2 hr	1.5 hr	1 hr	0.5 hr
1.78 Final VPC	M12V40(F)	1.7	3.4	4.0	4.3	4.9	5.5	6.3	7.5	9.2	11.8	13.9	16.8	21.3	30.2	50.7
	M12V70(F)	3.2	6.2	7.4	8.1	9.0	10.0	11.4	13.3	16.0	20.4	23.8	28.7	36.5	51.3	90.0
	M12V90(F)	4.1	7.7	9.1	10.0	11.2	12.6	14.2	16.6	20.0	25.5	29.8	36.0	45.9	64.7	105.2
	M6V190(F)	9.0	16.5	19.4	21.2	23.6	26.5	30.3	35.6	43.3	55.5	65.0	78.8	101.1	143.5	241.6

	Model Number	Time														
		24 hr	12 hr	10 hr	9 hr	8 hr	7 hr	6 hr	5 hr	4 hr	3 hr	2.5 hr	2 hr	1.5 hr	1 hr	0.5 hr
1.80 Final VPC	M12V40(F)	1.7	3.4	4.0	4.3	4.9	5.5	6.3	7.5	9.2	11.8	13.8	16.7	21.2	30.0	50.0
	M12V70(F)	3.2	6.2	7.4	8.0	8.9	9.9	11.3	13.2	15.9	20.3	23.6	28.5	36.3	51.0	89.0
	M12V90(F)	4.1	7.7	9.1	10.0	11.1	12.5	14.1	16.5	19.9	25.3	29.5	35.7	45.5	64.0	103.3
	M6V190(F)	8.9	16.4	19.2	21.1	23.4	26.3	30.1	35.3	43.0	55.0	64.3	77.9	99.9	142.0	236.5

	Model Number	Time														
		24 hr	12 hr	10 hr	9 hr	8 hr	7 hr	6 hr	5 hr	4 hr	3 hr	2.5 hr	2 hr	1.5 hr	1 hr	0.5 hr
1.81 Final VPC	M12V40(F)	1.7	3.4	4.0	4.3	4.8	5.4	6.2	7.4	9.1	11.7	13.7	16.6	21.1	29.9	49.6
	M12V70(F)	3.1	6.1	7.3	8.0	8.8	9.9	11.2	13.1	15.8	20.1	23.4	28.2	35.9	50.6	87.4
	M12V90(F)	4.1	7.6	9.0	9.9	11.0	12.4	14.0	16.3	19.7	25.1	29.3	35.4	45.1	63.5	101.8
	M6V190(F)	8.8	16.2	19.0	20.9	23.2	26.1	29.9	35.1	42.6	54.5	63.6	76.9	98.5	139.5	231.0

	Model Number	Time														
		24 hr	12 hr	10 hr	9 hr	8 hr	7 hr	6 hr	5 hr	4 hr	3 hr	2.5 hr	2 hr	1.5 hr	1 hr	0.5 hr
1.83 Final VPC	M12V40(F)	1.6	3.3	3.9	4.2	4.7	5.3	6.1	7.3	9.0	11.6	13.6	16.4	20.9	29.5	48.5
	M12V70(F)	3.1	6.0	7.1	7.9	8.8	9.8	11.1	12.9	15.6	19.9	23.2	27.9	35.6	50.1	85.6
	M12V90(F)	4.0	7.5	8.9	9.8	10.8	12.2	13.8	16.1	19.4	24.7	28.8	34.8	44.3	62.4	100.3
	M6V190(F)	8.6	16.0	18.8	20.7	23.0	25.9	29.6	34.8	42.2	53.7	62.7	75.8	96.8	136.5	224.3

MARATHON® Performance Specifications Amperes @ 25°C (77°F)

**1.85
Final
VPC**

Model Number	Time														
	24 hr	12 hr	10 hr	9 hr	8 hr	7 hr	6 hr	5 hr	4 hr	3 hr	2.5 hr	2 hr	1.5 hr	1 hr	0.5 hr
M12V40(F)	1.6	3.3	3.9	4.2	4.7	5.3	6.1	7.2	8.8	11.4	13.3	16.0	20.4	28.8	47.2
M12V70(F)	3.0	5.9	7.0	7.7	8.6	9.6	10.9	12.8	15.4	19.6	22.9	27.6	35.1	49.4	83.0
M12V90(F)	4.0	7.4	8.7	9.6	10.7	12.0	13.6	15.8	19.1	24.3	28.3	34.2	43.6	61.3	98.0
M6V190(F)	8.5	15.7	18.5	20.3	22.6	25.4	29.2	34.4	41.5	52.7	61.5	74.2	94.6	133.1	216.6

**1.87
Final
VPC**

Model Number	Time														
	24 hr	12 hr	10 hr	9 hr	8 hr	7 hr	6 hr	5 hr	4 hr	3 hr	2.5 hr	2 hr	1.5 hr	1 hr	0.5 hr
M12V40(F)	1.6	3.1	3.6	4.0	4.5	5.1	5.8	6.9	8.4	10.9	12.7	15.4	19.5	27.3	44.6
M12V70(F)	3.0	5.7	6.8	7.5	8.3	9.3	10.6	12.4	14.9	18.9	22.0	26.6	33.8	47.4	78.6
M12V90(F)	3.9	7.2	8.5	9.3	10.3	11.6	13.1	15.3	18.5	23.5	27.4	33.0	41.9	58.8	93.7
M6V190(F)	8.2	15.2	17.9	19.7	21.8	24.6	28.2	33.1	40.0	50.8	59.2	71.3	90.7	127.3	204.3

**1.90
Final
VPC**

Model Number	Time														
	24 hr	12 hr	10 hr	9 hr	8 hr	7 hr	6 hr	5 hr	4 hr	3 hr	2.5 hr	2 hr	1.5 hr	1 hr	0.5 hr
M12V30T	1.1	2.2	2.6	2.8	3.1	3.6	4.1	4.7	5.7	7.2	8.4	10.2	13.0	18.3	31.0
M12V40(F)	1.5	2.9	3.4	3.8	4.2	4.8	5.6	6.6	8.0	10.4	12.2	14.7	18.6	25.9	42.4
M12V70(F)	2.8	5.4	6.4	7.1	7.9	9.0	10.1	11.7	14.1	17.9	20.8	25.1	31.8	44.4	72.5
M12V90(F)	3.7	6.8	8.0	8.8	9.8	11.0	12.6	14.6	17.5	22.2	25.8	31.1	39.5	55.2	87.6
M6V190(F)	7.8	14.5	17.0	18.7	20.8	23.4	26.8	31.5	37.8	47.9	55.7	67.0	84.9	118.7	187.8

**1.92
Final
VPC**

Model Number	Time														
	24 hr	12 hr	10 hr	9 hr	8 hr	7 hr	6 hr	5 hr	4 hr	3 hr	2.5 hr	2 hr	1.5 hr	1 hr	0.5 hr
M12V40(F)	1.4	2.8	3.2	3.6	4.0	4.5	5.2	6.2	7.6	9.8	11.5	13.9	17.6	24.4	39.8
M12V70(F)	2.7	5.1	6.1	6.7	7.4	8.4	9.5	11.1	13.3	16.8	19.5	23.5	29.7	41.4	64.6
M12V90(F)	3.6	6.5	7.6	8.3	9.2	10.3	11.8	13.8	16.7	21.2	24.6	29.6	37.2	51.4	81.9
M6V190(F)	7.5	13.7	16.1	17.7	19.6	22.1	25.3	29.7	36.0	45.1	52.3	62.8	79.4	110.4	146.8

**1.94
Final
VPC**

Model Number	Time														
	24 hr	12 hr	10 hr	9 hr	8 hr	7 hr	6 hr	5 hr	4 hr	3 hr	2.5 hr	2 hr	1.5 hr	1 hr	0.5 hr
M12V40(F)	1.3	2.5	3.0	3.3	3.7	4.2	4.9	5.7	7.0	9.1	10.7	12.9	16.3	22.5	36.7
M12V70(F)	2.5	4.8	5.7	6.3	7.0	8.0	9.0	10.4	12.5	15.8	18.3	21.9	27.6	38.4	54.9
M12V90(F)	3.3	6.1	7.1	7.8	8.7	9.7	11.1	13.0	15.8	20.2	23.5	28.2	35.1	47.8	76.4
M6V190(F)	7.0	13.0	15.3	16.7	18.6	20.9	23.9	28.1	33.5	42.3	49.0	58.6	73.8	102.3	128.6

MARATHON® Performance Specifications Watts per Cell @ 25°C (77°F)

**1.75
Final
VPC**

Model Number	Time											
	24 hr	12 hr	10 hr	9 hr	8 hr	6 hr	4 hr	3 hr	2 hr	1.5 hr	1 hr	0.5 hr
M12V40(F)	2.9	5.5	6.6	7.2	8.1	10.6	15.4	20.1	29.2	38.0	54.6	96.5
M12V70(F)	6.7	12.6	14.9	16.4	18.3	23.2	32.3	40.8	56.8	71.8	100.0	174.0
M12V90(F)	7.7	14.9	17.7	19.6	21.9	27.7	39.3	50.3	71.2	91.0	128.7	209.0
M6V190(F)	17.8	33.3	39.2	43.2	48.0	62.4	90.0	112.4	155.8	197.1	274.7	465.0

**1.78
Final
VPC**

Model Number	Time											
	24 hr	12 hr	10 hr	9 hr	8 hr	6 hr	4 hr	3 hr	2 hr	1.5 hr	1 hr	0.5 hr
M12V40(F)	2.8	5.5	6.5	7.2	8.1	10.5	15.3	20.0	29.1	37.8	54.3	96.0
M12V70(F)	6.7	12.6	14.8	16.3	18.2	23.0	32.1	40.6	56.5	71.5	99.5	172.0
M12V90(F)	7.6	14.8	17.5	19.4	21.7	27.5	38.9	49.8	70.4	90.0	127.3	206.0
M6V190(F)	17.6	33.0	38.9	42.8	47.6	61.8	89.3	111.1	154.5	195.4	272.0	456.1

**1.80
Final
VPC**

Model Number	Time											
	24 hr	12 hr	10 hr	9 hr	8 hr	6 hr	4 hr	3 hr	2 hr	1.5 hr	1 hr	0.5 hr
M12V40(F)	2.8	5.5	6.5	7.2	8.0	10.5	15.3	19.9	29.0	37.6	54.0	95.1
M12V70(F)	6.7	12.5	14.8	16.2	18.1	23.0	31.9	40.4	56.2	71.0	98.9	170.0
M12V90(F)	7.6	14.7	17.4	19.3	21.6	27.3	38.6	49.4	69.8	89.2	126.2	203.1
M6V190(F)	17.5	32.8	38.7	42.5	47.3	61.4	88.7	110.4	153.4	193.7	269.0	447.1

**1.83
Final
VPC**

Model Number	Time											
	24 hr	12 hr	10 hr	9 hr	8 hr	6 hr	4 hr	3 hr	2 hr	1.5 hr	1 hr	0.5 hr
M12V40(F)	2.8	5.5	6.5	7.2	8.0	10.4	15.2	19.8	28.7	37.3	53.3	93.5
M12V70(F)	6.5	12.2	14.4	15.9	17.7	22.6	31.3	39.5	55.0	69.6	96.9	164.0
M12V90(F)	7.4	14.3	17.1	18.9	21.1	26.8	37.7	48.2	67.9	86.7	122.2	195.1
M6V190(F)	17.2	32.1	37.8	41.6	46.2	59.9	86.2	108.5	151.0	189.2	260.0	424.4

MARATHON® Performance Specifications Watts per Cell @ 25°C (77°F)

1.85 Final VPC	Model Number	Time											
		24 hr	12 hr	10 hr	9 hr	8 hr	6 hr	4 hr	3 hr	2 hr	1.5 hr	1 hr	0.5 hr
	M12V40(F)	2.8	5.4	6.5	7.1	8.0	10.4	15.1	19.7	28.5	37.0	52.8	92.0
	M12V70(F)	6.5	12.1	14.2	15.7	17.4	22.3	30.9	39.0	54.1	68.5	95.4	160.0
	M12V90(F)	7.3	14.1	16.8	18.6	20.8	26.4	37.1	47.3	66.6	84.9	119.6	190.0
	M6V190(F)	16.9	31.7	37.4	41.1	45.8	59.3	85.0	107.0	149.0	186.2	254.9	410.1

1.87 Final VPC	Model Number	Time											
		24 hr	12 hr	10 hr	9 hr	8 hr	6 hr	4 hr	3 hr	2 hr	1.5 hr	1 hr	0.5 hr
	M12V40(F)	2.8	5.4	6.4	7.1	7.9	10.3	14.9	19.4	28.1	36.3	51.4	89.3
	M12V70(F)	6.3	11.7	13.8	15.2	16.9	21.5	30.0	37.9	52.9	66.9	93.3	154.9
	M12V90(F)	7.1	13.7	16.2	17.9	20.1	25.6	36.0	45.8	64.3	81.9	115.1	181.7
	M6V190(F)	16.6	30.8	36.2	39.7	44.1	56.8	81.4	105.0	145.4	180.6	245.0	388.3

1.90 Final VPC	Model Number	Time											
		24 hr	12 hr	10 hr	9 hr	8 hr	6 hr	4 hr	3 hr	2 hr	1.5 hr	1 hr	0.5 hr
	M12V40(F)	2.7	5.2	6.2	6.9	7.6	10.0	14.4	16.7	26.8	34.5	48.7	84.2
	M12V70(F)	6.0	11.2	13.2	14.5	16.1	20.8	28.6	36.1	50.2	63.4	88.1	144.6
	M12V90(F)	6.8	13.0	15.4	17.0	19.0	24.4	34.2	43.5	60.9	77.4	108.4	169.9
	M6V190(F)	16.1	29.5	34.6	37.9	42.0	54.1	77.1	100.0	140.0	172.4	231.3	358.1

1.92 Final VPC	Model Number	Time											
		24 hr	12 hr	10 hr	9 hr	8 hr	6 hr	4 hr	3 hr	2 hr	1.5 hr	1 hr	0.5 hr
	M12V40(F)	2.6	5.0	6.0	6.6	7.3	9.5	13.8	17.8	25.5	32.7	45.9	79.3
	M12V70(F)	5.7	10.6	12.5	13.7	15.2	19.7	27.1	34.2	47.4	59.7	82.7	134.7
	M12V90(F)	6.4	12.3	14.5	16.1	17.9	23.3	32.4	41.1	57.3	72.5	101.1	155.4
	M6V190(F)	15.4	28.2	33.0	36.2	40.1	51.5	73.4	95.0	132.0	162.4	217.4	296.7

1.94 Final VPC	Model Number	Time											
		24 hr	12 hr	10 hr	9 hr	8 hr	6 hr	4 hr	3 hr	2 hr	1.5 hr	1 hr	0.5 hr
	M12V40(F)	2.5	4.8	5.6	6.2	6.9	9.0	12.9	16.7	23.9	30.6	42.8	73.7
	M12V70(F)	5.4	10.0	11.8	12.9	14.4	18.6	25.7	32.3	44.6	56.1	77.3	125.5
	M12V90(F)	6.1	11.6	13.7	15.1	16.8	21.8	30.8	38.8	53.7	67.7	93.8	144.3
	M6V190(F)	14.6	26.8	31.4	34.5	38.2	49.2	70.3	90.5	125.2	153.4	204.1	262.8

GNB Industrial Power – The Industry Leader.



GNB Industrial Power, a division of Exide Technologies, is a global leader in network power applications including communication/data networks, UPS systems for computers and control systems, electrical power generation and distribution systems, as well as a wide range of other industrial standby power applications. With a strong manufacturing base in both North America and Europe and a truly global reach (operations in more than 80 countries) in sales and service, GNB Industrial Power is best positioned to satisfy your back up power needs locally as well as all over the world.

Based on over 100 years of technological innovation the Network Power group leads the industry with the most recognized global brands such as ABSOLYTE®, GNB® FLOODED CLASSIC®, MARATHON®, RELAY GEL®, SONNENSCHN® and SPRINTER®. They have come to symbolize quality, reliability, performance and excellence in all the markets served.

GNB Industrial Power takes pride in its commitment to a better environment. Its Total Battery Management program, an integrated approach to manufacturing, distributing and recycling of lead acid batteries, has been developed to ensure a safe and responsible life cycle for all of its products.

GNB Industrial Power

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